

SHORT ANSWER QUESTIONS

Answer the following questions in the space provided.

THINKING/INQUIRY (21 MARKS)

1. In a calorimetry experiment, 250.0 mL of 0.120 mol/L BaCl₂(aq) solution and 150.0 mL of 0.200 mol/L Na₂SO₄(aq) solution were mixed. A precipitate of BaSO₄(s) forms. The temperature rose from 20.0°C to 20.49°C. Assuming the density of all solutions is 1.00 g/mL and that their specific heat capacity is 4.180 J/g°C, calculate the enthalpy change of the reaction in kJ/mol of Na₂SO₄(aq). (4 marks)

$$250.0 \text{ mL} \approx 250 \text{ mL}$$

$$0.25 \text{ L} \cdot 0.120 \text{ mol/L} = 0.03 \text{ mol BaCl}_2$$

$$n_{\text{water}} = \Delta H_{\text{rxn}}$$

$$\Delta H_{\text{rxn}} = \Delta H_{\text{rxn}}$$

$$(0.49^\circ) = \Delta H_{\text{rxn}}$$

$$-3886.5 \text{ J/mol Na}_2\text{SO}_4$$

How many moles of Na₂SO₄?

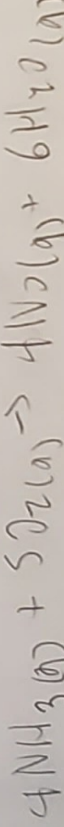
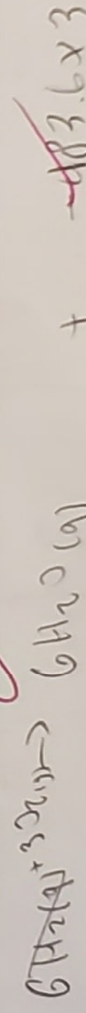
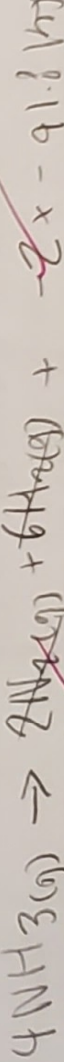
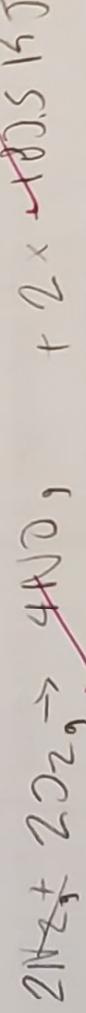
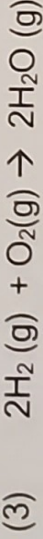
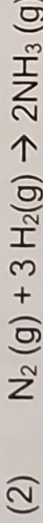
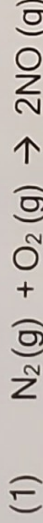
$$n_{\text{Na}_2\text{SO}_4} = \frac{150/1000 \text{ L} \cdot 0.2 \text{ mol/L}}{142.04 \text{ g/mol}}$$

$$n_{\text{Na}_2\text{SO}_4} = 0.00214 \text{ mol}$$

$$\eta = CV = 0.200 \text{ mol} \times 0.150 \text{ L}$$

$$= 0.03 \text{ mol}$$

2. Calculate the enthalpy change for the following reaction from the following equations: (4 marks)



$$\Delta H^\circ = (2 \times (-180.5)) + (2 \times (-91.8)) + (3 \times (-483.6))$$

$$\Delta H^\circ_{\text{rxn}} = -1628.2 \text{ kJ}$$

Using Hess's Law

3. The following data was collected for four trials of the following experiment:

Experiment	Initial [A] mol/L	Initial [B] mol/L	Initial [C] mol/L	$-\Delta[A]/\Delta t$ mol/L·s
1	0.10	0.20	0.30	9.0×10^{-5}
2	0.30	0.20	0.30	8.1×10^{-4}
3	0.10	0.30	0.30	9.0×10^{-5}
4	0.10	0.20	0.90	2.7×10^{-4}

- (a) Use this information to determine the order of the reactants. Show your work! (3 marks)

$$k[A]^a[B]^b[C]^c = \text{Rate}$$

$$\frac{\text{Rate}_2}{\text{Rate}_1} = \left(\frac{[A]_2}{[A]_1} \right)^a$$

$$9 = [3]^a \quad a=2$$

$$\frac{\text{Rate}_3}{\text{Rate}_2} = \frac{k[A]_3^a[B]_3^b[C]_3^c}{k[A]_2^a[B]_2^b[C]_2^c}$$

$$\frac{1}{9} = \left(\frac{[A]_3}{[A]_2} \right)^a \cdot \left(\frac{[B]_3}{[B]_2} \right)^b$$

$$\left(\frac{1}{3} \right)^2 = \left(\frac{1}{3} \right)^2 \cdot \left(\frac{[B]_3}{[B]_2} \right)^b \quad b=0$$

- (b) What is the value of the rate constant, k ? (2 marks)

$$\text{Rate} = k[A]^2[B]^0[C]^0$$

$$9 \times 10^{-5} = k[0.10]^2[0.30]^0$$

$$k = 3.0 \times 10^{-3} \text{ s}^{-1}$$

$$\text{Units } k = \text{M}^{-2} \text{ s}^{-1}$$

$$\frac{\text{M}}{\text{s}} = k \cdot \text{M} \cdot \text{M} \cdot \text{M}$$

$$\frac{\text{M}}{\text{s}} = k \cdot \text{M}^3 \quad k = \frac{\text{M}^{-2}}{\text{s}}$$

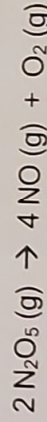
- (c) Write the rate law expression. (1 mark)

$$\text{Rate} = (0.03 \frac{\text{L}^2}{\text{mol}^2 \text{s}}) [A]^2 [C]^0$$

- (d) What is the overall order for the reaction? (1 mark)

$$3$$

4. Dinitrogen pentoxide decomposes to nitrogen dioxide and oxygen gas by a first-order reaction with a rate constant of $5.1 \times 10^{-4} \text{ s}^{-1}$ at 45°C .



- (a) If the initial concentration of N_2O_5 is 0.30 M , what is the concentration after 180 s ? (2 marks)

First order so

$$\ln[A_1] = \ln[A_0] - kt$$

$$\ln[A_1] = \ln(0.30 \text{ M}) - (5.1 \times 10^{-4} \text{ s}^{-1}) \cdot 180$$

$$A_1 = e^{\ln(0.30 \text{ M}) - (5.1 \times 10^{-4} \text{ s}^{-1}) \cdot 180}$$

$$A = 0.27 \text{ M after } 180 \text{ seconds}$$

(b) How long will it take (in seconds) for the concentration of N_2O_5 to decrease in half? (2 marks)

$$T_{1/2} = \frac{\ln(2)}{k} = \frac{\ln(2)}{5.1 \times 10^{-4} \text{ s}^{-1}}$$

$k \propto 1st \text{ order reaction}$

2 $T_{1/2} = 1359.12 \text{ s}$

(c) How long will it take to convert 80% of the starting material? (2 marks)

$$\ln[A] = \ln[A_0] - kt \quad A = 1, \cdot 0.20$$

2 $\ln\left[\frac{A}{A_0}\right] = -kt$

$$\frac{\ln(0.2)}{-5.1 \times 10^{-4} \text{ s}^{-1}} = +$$

$$+ = 3155.8 \text{ seconds to convert 80\% of material}$$

or have 20% remaining $[A]$

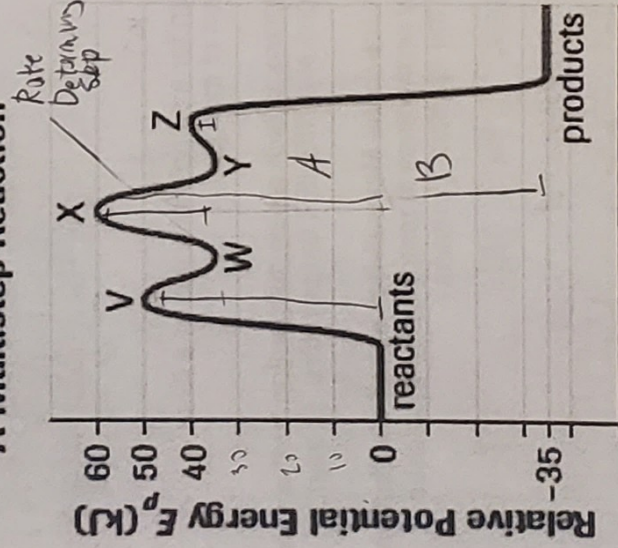
COMMUNICATION (12 MARKS)

1. Examine the following Potential Energy Diagram and answer the following questions.

a) Is this reaction below exothermic or endothermic? (1 mark) exothermic

b) Using the graph, answer the following questions. Label with arrows and letters where appropriate. (3 marks)

A Multistep Reaction

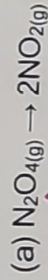


What is the numerical value of the overall activation energy for all the steps in the reaction? Label on the graph with the letter A.	$E_a = 62 \text{ KJ}$
What is the numerical value of the enthalpy change for the reaction (ΔH)? Label on the graph with the letter B.	$\Delta H = -35 \text{ KJ}$
Which step is the rate determining step?	From $W \rightarrow X \rightarrow Y$

Right

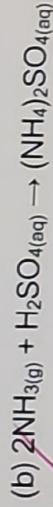
Reaction Progress

2. For each of the following reactions, indicate whether you would expect the entropy of the system to increase or decrease, and explain why. (4 marks)



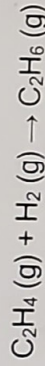
Increase ✓ as there are more moles of gas on the reaction is complete (states are the same)

4



Decrease as there are less moles on the right of gas and gas $\rightarrow$ liquid is a decrease in entropy.

3. The hydrogenation of ethene gas at 25°C shows a decrease in disorder ($\Delta S^\circ = -120.7 \text{ J}/(\text{mol}\cdot\text{K})$) during an exothermic reaction. Determine whether the reaction is spontaneous or nonspontaneous by calculating ΔG° . (4 marks)



$\Delta G = \Delta H - \Delta S \cdot T$

Now need to find ΔH°

$\Delta H = \sum \Delta H_{\text{products}} - \sum \Delta H_{\text{reactants}}$

$= (\Delta H_{\text{C}_2\text{H}_6}) - (\Delta H_{\text{C}_2\text{H}_4} + \Delta H_{\text{H}_2})$

$= (-83.8 \text{ kJ}) - (-52.5)$

$= -31.3 \text{ kJ}$

$\Delta G = -31.3 \text{ kJ} - \left(\frac{-120.7}{1000} \right) \cdot (273.15 + 25^\circ\text{C})$

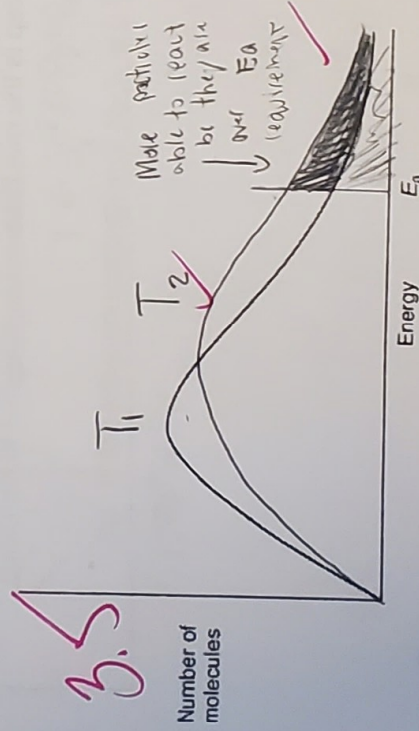
APPLICATION (10 MARKS)

$\Delta G = 4.6817$

-100.33 kJ

Spontaneous in reverse as ΔG is positive, so nonspontaneous in forward direction.

1. The diagram below shows the Maxwell-Boltzmann distribution for a sample of a gas at a fixed temperature. E_a is the activation energy for the decomposition of this gas.



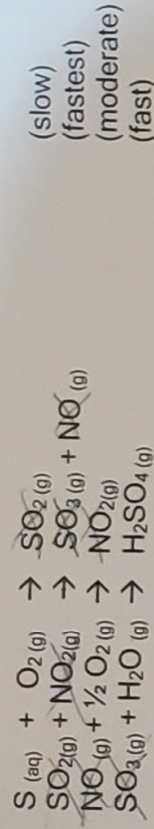
- (a) On this diagram, sketch a new distribution for the same sample of gas at a HIGHER temperature. (2 marks)

- (b) With reference to the Maxwell-Boltzmann distribution, explain why an increase in temperature increases the rate of the reaction. (2 marks)

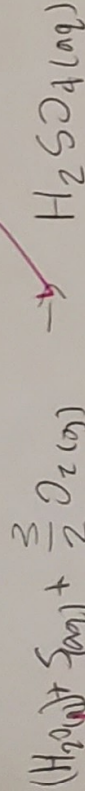
It increases the value as the average energy of each particle is shifted higher. This means a greater % will be over the

activation energy leading to more reactions occurring by probability.
↑ effective collisions

2. A chemist proposes the following reaction mechanism:



(a) Write the overall reaction equation. (1 mark)



(b) List any reaction intermediate(s) and/or catalyst(s). (2 marks)

Intermediate(s): $\text{SO}_{2(g)}, \text{SO}_{3(g)}, \text{NO}_{(g)}$ Catalyst(s): $\text{NO}_{2(g)}$

(c) The rate law equation was determined to be $\text{rate} = k[\text{S}][\text{O}_2]^2$. Is the proposed mechanism plausible? Support your answer. (1 mark)

Using in given steps $k[\text{S}][\text{O}_2]$ is the rate because $\text{S} + \text{O} \rightarrow \text{SO}_2$ is the slowest step and its rate is $k[\text{S}][\text{O}_2]$

(d) A new catalyst is discovered that decreases the activation energy for step 2. How will this affect the overall rate of the reaction? Give a reason for your answer. (2 marks)

This won't affect the overall rate of the reaction. This is because the rate of a reaction is dependent on the slowest step (1st), so making step 2 faster by decreasing E_a (threshold for reaction, lowering E_a by rate $\uparrow$) does nothing to step 1.